

5.

Ans: C

relative speed of approach = relative speed of separation

$$u_A - u_B = v_B - v_A$$

$$1.2 - 0 = v_B - v_A$$

By conservation of momentum

$$m_A u_A + m_B u_B = m_A v_A + m_B v_B$$

$$m(1.2) + 0 = mv_A + mv_B$$

$$1.2 = v_A + v_B$$

$$v_B = 1.2$$

$$v_A = 0$$

Note: For the elastic collision, when the two masses are identical, they swap velocities after collision.

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Ans: A